Task 29: Final Summary Report

Best-Performing Model and Why

- **Multiple Linear Regression (MLR)** performed best overall.
- It balanced **bias and variance**, explained more variance in house prices (higher R^2, Adjusted R^2), and reduced prediction errors compared to Simple LR.
- Polynomial Regression improved fit but risked **overfitting**, making MLR the most reliable choice.

Impact of Gradient Descent Optimization

- Gradient Descent methods (Batch, SGD, Mini-Batch) ensured efficient parameter estimation.
- **Batch GD** was stable but slow.
- **SGD** was fast but noisy.
- **Mini-Batch GD** provided the best balance of speed and stability, making it the practical choice for larger datasets.

Evidence of Overfitting/Underfitting

- **Simple LR:** Underfitting due to high bias (too simplistic).
- **Polynomial Regression (higher degree):** Overfitting due to high variance (too complex).
- **Multiple LR:** Achieved the best trade-off, avoiding both extremes.

Practical Business Interpretation

- House prices are influenced by multiple factors: area, bedrooms, bathrooms, location, age, and lot size.
- **Area_sqft** alone is not sufficient; location and amenities significantly impact pricing.
- A Multiple LR model provides actionable insights for real estate pricing, investment decisions, and customer guidance.
- Gradient Descent optimization ensures scalability for larger housing datasets.